PUBUDU GUNASEKARA

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Master of Science, Northeastern University | Seeking Software Engineering Internship, Summer 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences, San Jose, CA 01/2026 - 05/2028 (Expected)
Master of Science in Computer Science | GPA: 4.0 / 4.0
Coursework: Algorithms, Programming Design Paradigms, Machine Learning, Natural Language Processing

Conestoga College, Postgraduate Certificate in Mobile Solutions Development, GPA 3.74 / 4.0 2022 - 2023

Victoria University, Bachelor of Information Technology, GPA 6.25 / 7.0 2018 - 2021

EXPERIENCE

Food Services & Hospitality, Team Member, Waterloo, Canada 01/2024 - 12/2025

Virtusa, Associate Engineer (Quality Assurance), Colombo, Sri Lanka 06/2021 - 03/2022

- Wrote **Java and Selenium WebDriver** tests and ran test plans in agile sprints for a British Telecommunications (BT) platform serving **1M+ users**.
- Closed about **10 tickets per 2-week sprint**, 2-3 days early, then handled peer review, ad hoc tasks, client demos and meetings. Tracked defect density and open defect counts.

PROJECTS

Distributed Task Scheduler Case Study github.com/PubuduGunasekara/distributed-task-scheduler
Java 21, Spring Boot, Kafka, Redis, PostgreSQL, Prometheus, Docker, GitHub Actions

- Task scheduler on **hexagonal architecture** (ArchUnit-enforced). Prevents duplicate execution under Kafka's at-least-once delivery with a database state machine, optimistic locking and a Redis lock. Failures: 3 retries (10s, 30s, 90s), dead-letter topic, recovery job for stuck tasks.
- **188 tests**, **93.4% line and 91.1% branch coverage** (80% CI gate). Load test throughput: **~2,700 tasks/min** (1,000 simulated tasks, 3 consumers, local), zero duplicate runs.

AI Code Review Assistant Live Demo github.com/PubuduGunasekara/ai-code-reviewer
React 19/Vite, Node.js/Express, PostgreSQL, Redis, gpt-4o-mini, Docker, AWS

- GitHub OAuth app (React on AWS Amplify, Express on EC2) that reviews pull requests. Each finding has a severity, category, CWE reference and suggested fix.
- Strict JSON-schema output (OpenAI Structured Outputs): an invalid response fails the review instead of returning made-up results. Hourly per-user rate limit (Lua), Redis cache, 19 tests. 5 test reviews: **8.4 s** latency, about **$0.002** each.

Smart Farm Demo github.com/PubuduGunasekara/smart-farm-1.1.0
React Native (Expo), Redux, Firebase, Arduino, ESP

- Built the app's **device-control section** and the physical Arduino farm model. The app sends water, feed, gate and cleaning commands through Firebase to WiFi Arduino/ESP controllers that drive relays and motors.
- **Top 10 Most Innovative Projects** at the NSBM Green EXE v1.0 software competition (team project).

Travel Day Scheduler github.com/PubuduGunasekara/SmartTravelPlanner
Python, Weighted A Search*

- Team of 2, graduate algorithms: day-trip planner using A* search. Wrote the first search and the route map.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C++ (Arduino)
Backend: Spring Boot, Node.js, Express, Kafka, Redis, PostgreSQL, MongoDB, REST APIs
Infra & DevOps: Linux, Git, Docker, GitHub Actions, CI/CD, Jenkins, Prometheus, Grafana, AWS
Frontend & Mobile: React, Next.js, React Native, Vite, Redux, Firebase
Concepts: Distributed Systems, Data Structures, Object-Oriented Design, Concurrency, Unit Testing, Debugging
Certifications: Microsoft Azure Fundamentals (AZ-900), IBM Full Stack Software Developer